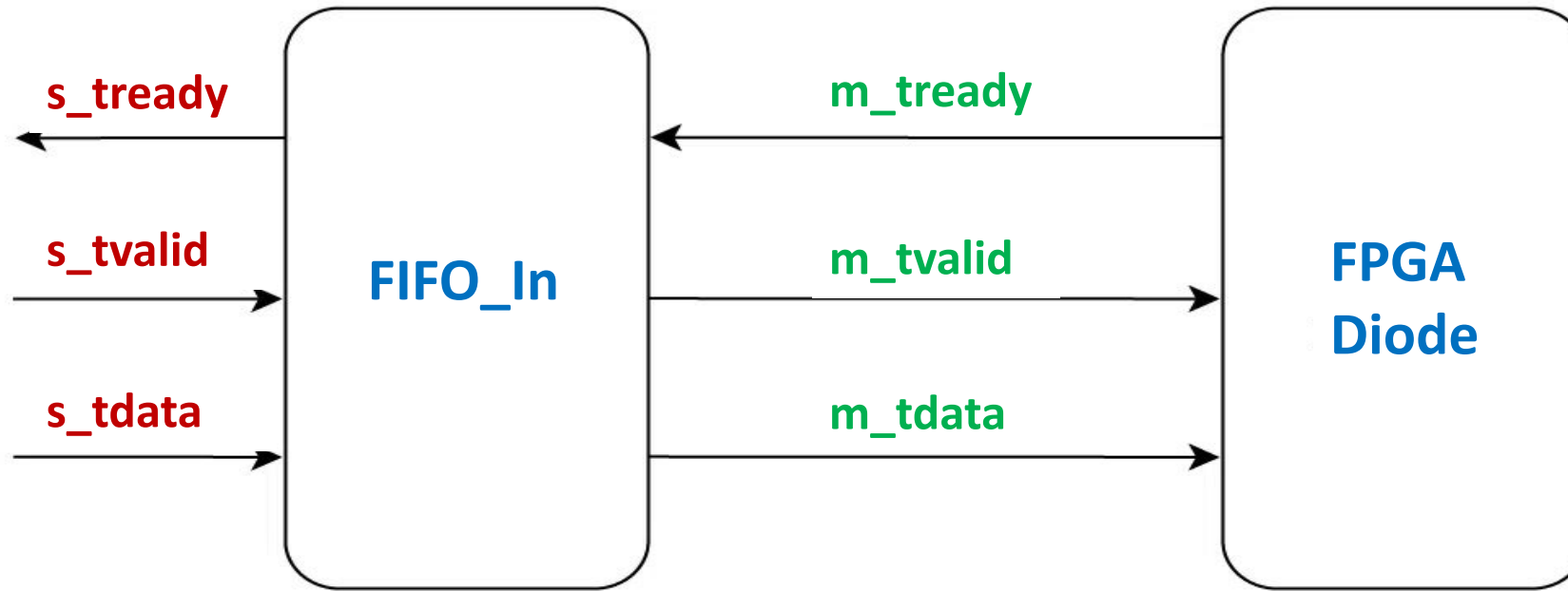


PROTOCOL: AXI-STREAM



s_tready Always set to 1 as there should be no backpressure on OT network.

s_tvalid This signals if the FIFO is ready to be written or not

s_tdata Contains DATA USER and LAST bits. It is the data generated from OT side

m_tready Signals if the FPGA diode is ready to receive data

m_tvalid Signals if the FIFO has valid data to be read

m_tdata Data propagated from OT Side to the subsequent stages

PACKET ATOMICITY VS BEAT LOGIC

- AXI Stream sees data as either words or packets.
- If the AXI stream buffers packets until tlast arrives and then transfers data as to ensure that partial packets are not formed, then the system is packet-atomic
- If the AXI stream stores bits as beats and allows data even if partial packets can be formed then it is said to be beat logic.
- We chose a hybrid model where packets are detected to be partial (corrupted) and then dropped instead of storing till the FIFO empties and new data can be written.

OVERFLOW AND UNDER LOW CONDITIONS

Overflow:

If the FIFO is full but the OT network fires write operation then FIFO is said to be in overflow.

wire overflow_cond = write_fire && fifo_full

Underflow:

- If the FIFO is empty but the IT network fires read operation then FIFO is said to be in underflow.
- However If FIFO is empty m_tvalid is 0. Only if m_tvalid is 1 IT network fires read operation.
- Therefore, it is highly unlikely for underflow to occur. Therefore we use starvation packet detection to detect packets where IT network is ready but FIFO is not able to provide data

wire underflow_cond = m_tready && !m_tvalid

PACKET CORRUPTION

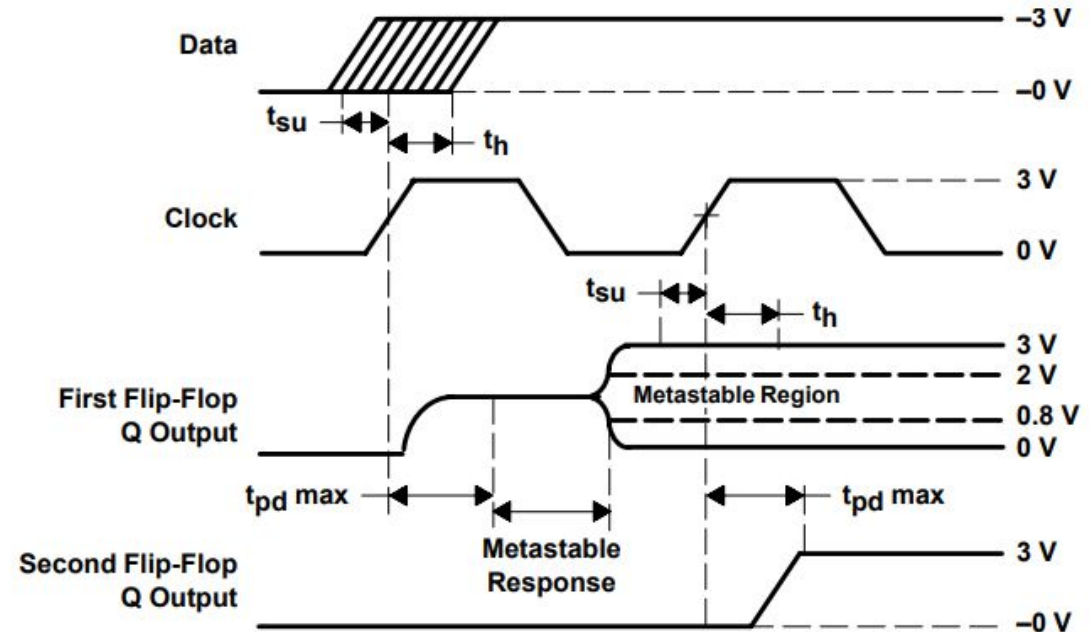
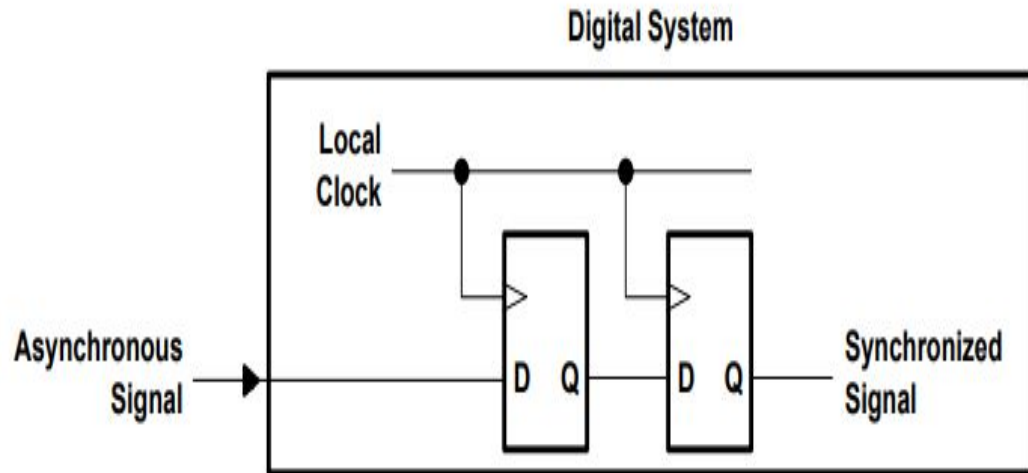
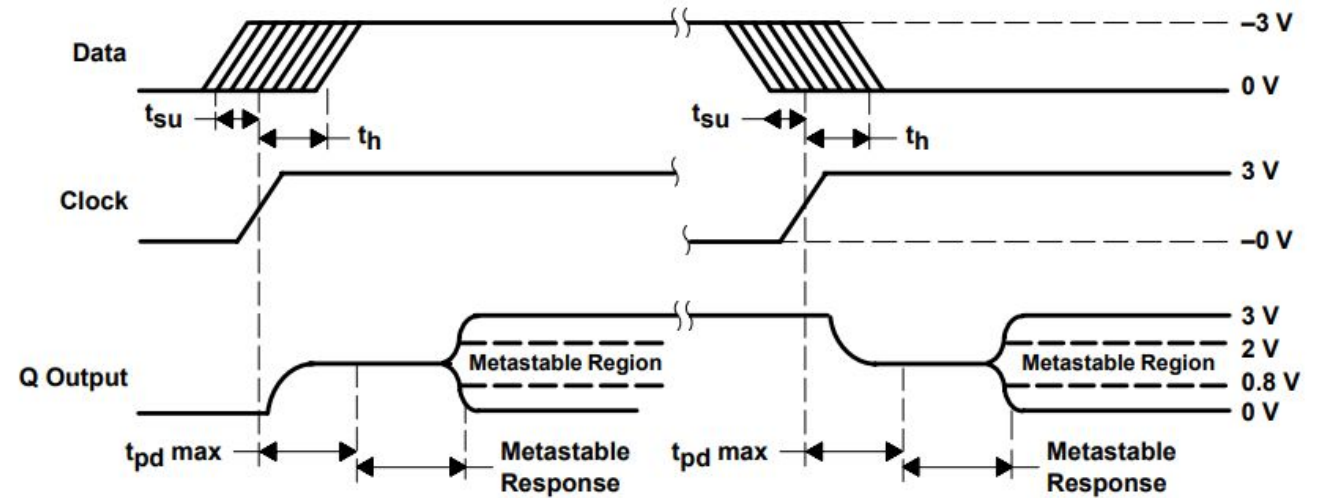
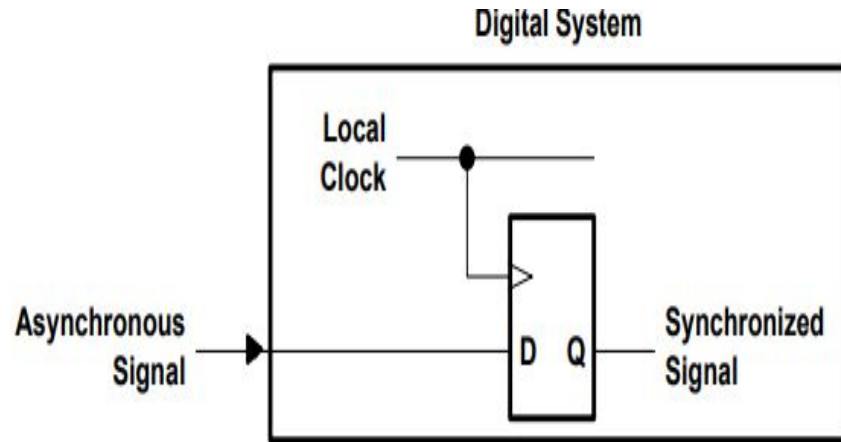
- Packet corruption occurs when an incomplete packet is transmitted to the parser.
- We use packet corruption to detect partial packets in a packet atomic system.
- However, in a packet atomic system we wait for the packet to completely fill and if we don't have space to transmit, the packet is rolled back.
- In this architecture, we prefer dropping the packet to ensure zero back pressure.

CDC MONITORING

- CDC monitoring involves timing analysis to check how stable a signal is moving from one clock domain to another. To emulate this, we use 2 clocks – clk and clk_monitor
- We use overflow and underflow syncs and monitors to identify the lagging and leading clocks and sync pulses accordingly.
- We use two clocks to ensure that metastability is almost zero. 2nd flipflop can fix the metastability caused by the 1st flipflop
- We use gray code to realise these flipflops because in gray code only 1 bit change occurs between successive increments.

CDC MONITORING

Source: ti.com



Comparison of 1 vs 2 flops for synchronisation

FORMAL ASSUMPTIONS

Formal assumptions (SVA) are constraints that help guide the debugger differentiate from valid and invalid test cases.

Examples:

- Assume that rst doesn't stick and becomes 0 eventually.
- Assume that count of a FIFO is not negative.

```
assert(count <= DEPTH);  
assert(count >= 0);  
assert(fifo_empty == (count == 0));  
assert(fifo_full == (count == DEPTH));
```

COMPLETE ARCHITECTURE

